Quantum Hackathon 2026 QML Project Report

Group Name:

Quantum Non-Metalheads

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Submission Category:

Rising Star - Quantum Machine Learning

Summary:

- Created scatter plot dividing elements into metals, non-metals, and metalloids.
- Created training data and testing data from element list.
- Created quantum kernel graphs from element list.
- Calculated the classical SVM and quantum SVC accuracies.
- Began evaluating and graphing test accuracy differences between classical SVM and quantum SVC.

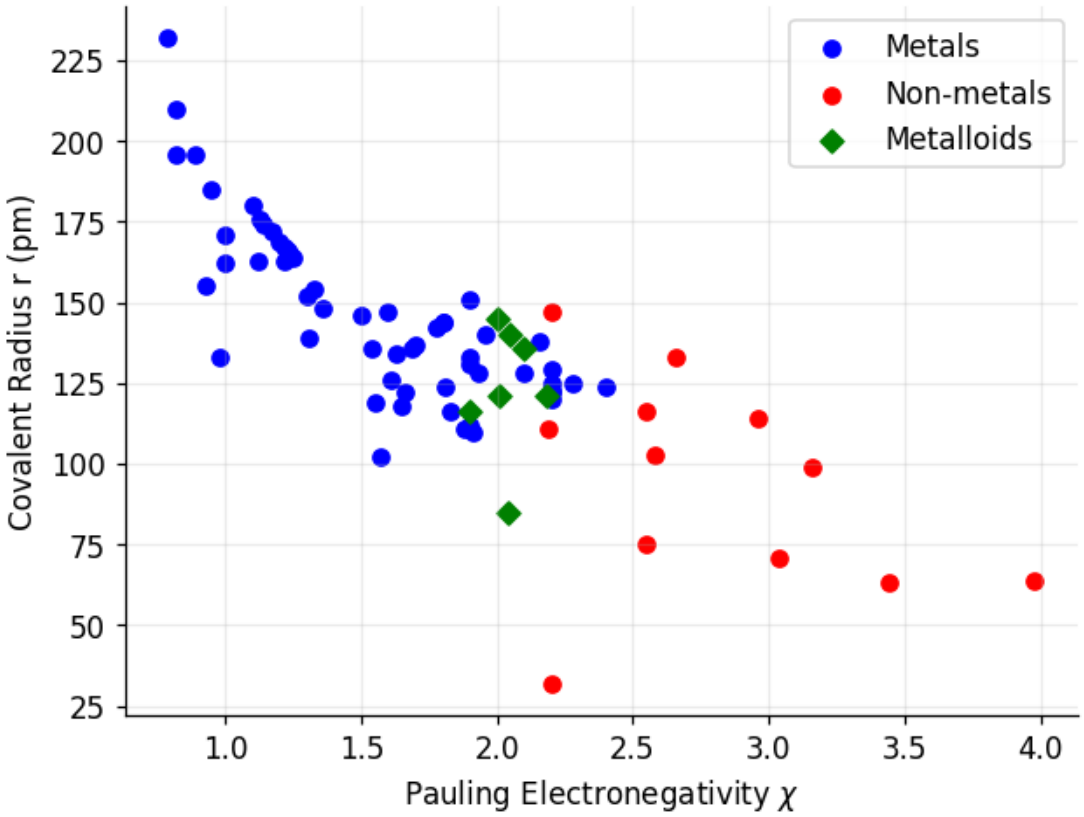
Motivation:

A desire to gain a better understanding of how quantum machine learning works and how quantum machine learning algorithms are implemented to solve problems. Interest in how quantum machine learning will affect current tools such as AI in the future as its usage and availability grows.

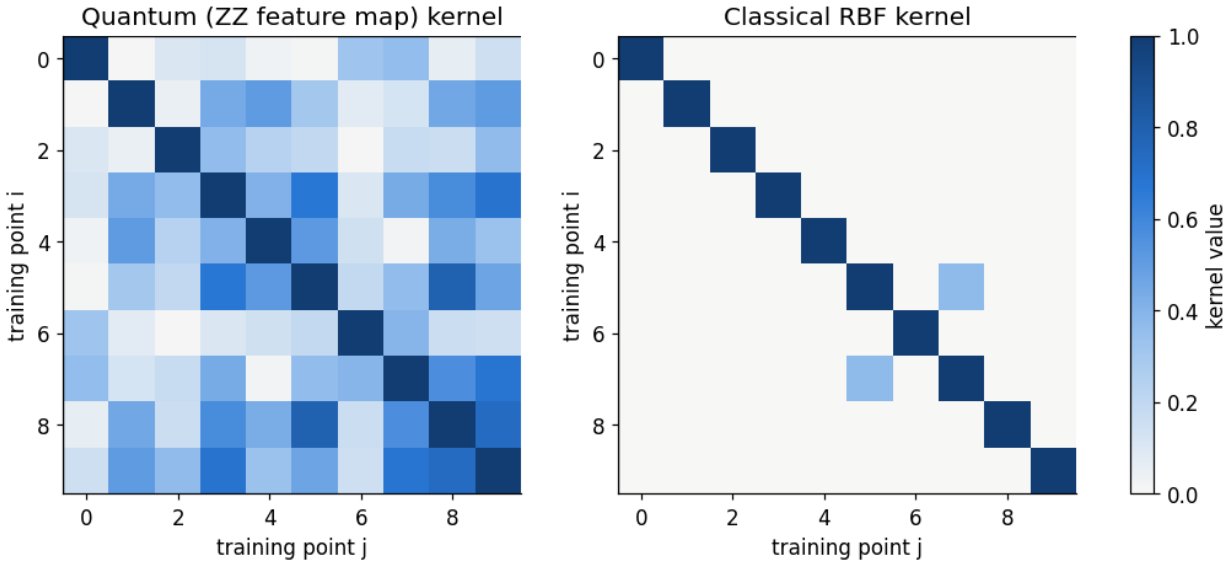
Code workflow:

- The first section of code installs all dependencies and libraries necessary for the code.
- The second section takes the database and extracts the electronegativity, radius, and class of each element in the data set. Then, it plots this information on a scatter plot.
- The next section sifts through the extracted information and creates training data and testing data.
- The next section goes through the data sets and compares the test accuracy of the SVC and QSVC components.
- The next section begins running the IBM instance and computes the ansatz.
- The final section looks at the VQC training and compares accuracies.

Results:



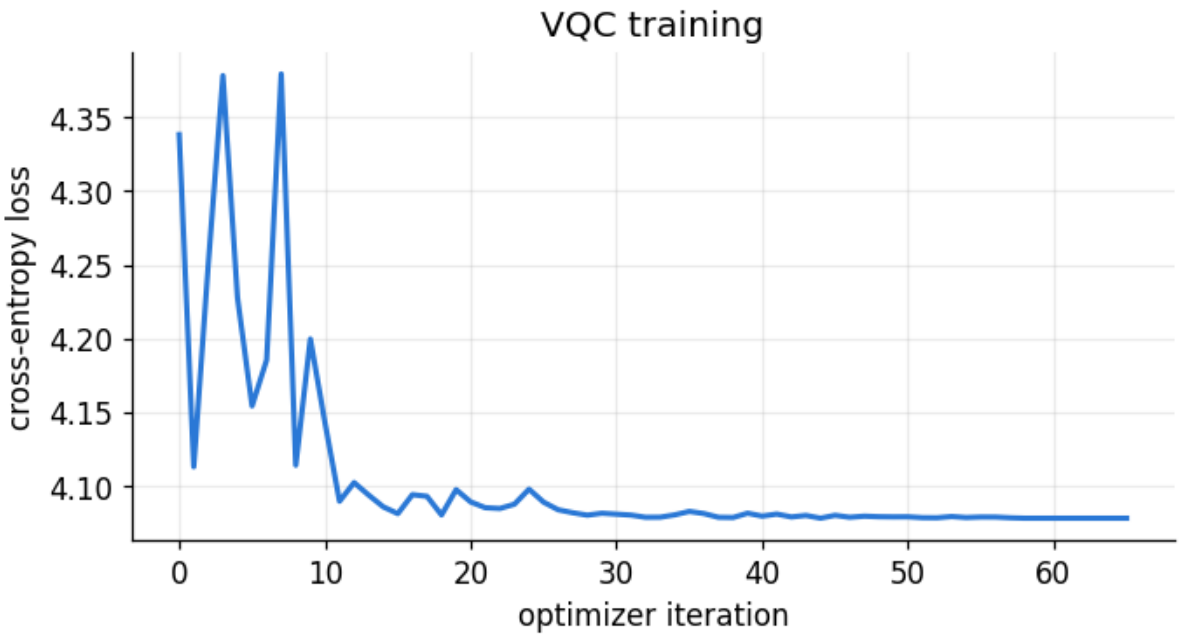
The Radius vs Electronegativity graph mapped the elements from the given data set. The different colors and shapes represent the class of the element described in the legend.



The next result that we obtained was for the kernels between the quantum and classical cases. The block diagonalization was found for both the quantum Kernel & classical Kernel. For our Kernel graphs, you can see that there is some small amount of block diagonalization appearing within the quantum kernel. When both $i, j > 7$, the map gets darker and more uniform. Whereas for the classical kernel there is almost no block diagonalization at all. Because it is lacking in that realm, it is more likely that our quantum kernel is sufficiently better for this run.

model	test accuracy	accuracy on the 65 extra points
SVC + quantum kernel (precomputed)	0.90	0.74
QSVC (Qiskit ML)	0.90	0.74
classical RBF SVM, tuned {'C': 0.1, 'gamma': 0.01}	0.90	0.74

To verify this, we compared mutliple models, starting with SVC + quantum kernel, QSVC, & classical RBF SVM. The test accuracy was the same among all three of the calculations, which implies that the quantum computations were more accurate than not.



model	test accuracy
VQC (ZZ map + RealAmplitudes, COBYLA)	0.20
quantum-kernel SVM (Part 4)	0.90
tuned RBF SVM (Part 4)	0.90

The last result from our project was a VQC training graph that shows the decrease in training loss as the number of iterations increase. Before 10 iterations, the cross-entropy loss was very high, and after 10, the loss dramatically decreases and stabilizes. We also gathered data to compare the accuracy of our VQC model with our quantum-kernel SVM & RBF SVM models. The VQC was unfortunately very low in comparison with the SVM test accuracies. This could indicate that the purely quantum approach might not have yielded the most accurate results for machine learning, or it could indicate that we had some issue calculating the VQC correctly in our code.